MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCFSC202 – FORENSIC BIOLOGY

Duration: 1.5 hrs

Maximum Marks: 50

Part A: Short Answer Questions

Answer any 10. Each carries 2 marks.

1. **Divisions of Forensic Biology:** Forensic Serology, Forensic Botany, Forensic Entomology, and DNA Profiling (or Wildlife Forensics/Forensic Anthropology).
2. **Amylase Detection in Saliva:** Amylase is a primary enzyme in saliva; its detection serves as a presumptive test to identify the presence of saliva on evidence like bite marks or cigarette butts.
3. **Phenolphthalein (Kastle-Meyer) Test:** A positive result is indicated by a **pink** color change.
4. **Acid Digestion for Diatoms:** Treating tissue (usually bone marrow or lung) with strong acids (e.g., Nitric Acid) to dissolve organic matter, leaving the silica-based diatom shells intact for microscopic analysis.
5. **Microscopic Fibre Cross-Sections:** Fibers are embedded in a medium (like paraffin) and sliced thinly to observe shapes (e.g., trilobal, circular) which help identify the fiber type and manufacturing process.
6. **Medullary Index:** The ratio of the diameter of the medulla to the diameter of the hair shaft. In humans, it is usually less than $1/3$; in animals, it is typically $1/2$ or greater.
7. **Stringing Method:** Using strings and protractors to trace the flight paths of blood droplets back to a common 3D space to find the exact location of the blow.
8. **Benzidine Test:** No longer preferred because Benzidine is a known **carcinogen**.
9. **Biochemical Composition of Hair:** Primarily composed of the protein **keratin**, along with lipids, water, and pigments (melanin).
10. **Confirmatory Test for Semen:** Identification of **Spermatozoa** via microscopy (Christmas Tree Stain) or the **PSA (p30)** test.
11. **Convergence vs. Origin:** Point of Convergence is the 2D intersection of blood drop paths on a surface; Point of Origin is the 3D location in space where the blood source was located.
12. **Diatom:** Unicellular algae with a unique cell wall made of silica (frustule) used in the diagnosis of drowning.

Part B: Short Essay Questions

Answer any 5. Each carries 4 marks.

13. Collection of Biological Evidence: Must emphasize preventing contamination and degradation. Use of sterile swabs, air-drying samples before packaging, and using paper bags (not plastic).

14. Acid Phosphatase (AP) Test: Semen contains high levels of AP. The test uses a substrate that turns purple in the presence of AP; it is a presumptive test for seminal stains.

15. Diatom Extraction from Lungs: Detailed steps involving sampling peripheral lung tissue, acid digestion, centrifugation, and slide preparation.

16. Fibre Comparison: Comparing color, diameter, delustrants, and optical properties (like birefringence) between a crime scene fiber and a known source.

17. Diatoms vs. Stomach Content: Diatoms in closed circulatory systems (bone marrow/brain) indicate the person was alive and breathing in the water, whereas stomach water can be swallowed post-mortem.

18. Properties of Blood: Discussion of **Surface Tension** (maintaining spherical shape), **Viscosity** (resistance to flow), and **Specific Gravity**.

19. Biochemical Basis of Blood Identification: Based on the **peroxidase-like activity** of hemoglobin, which catalyzes the oxidation of a reagent to produce a color change or luminescence.

Part C: Essay Questions

Answer any 1. Carries 10 marks.

20. Saliva, Sweat, and Faeces:

Significance: DNA source and linking suspect to scene (e.g., masks, clothing).

Collection: Swabbing for saliva/sweat; bulk collection for faecal matter.

Preservation: Dry storage and refrigeration to prevent bacterial growth.

21. Human vs. Animal Hair Comparison:

Cuticle: Human scales are imbricate; animals may have coronal or spinous scales.

Medulla: Human medulla is often absent, fragmented, or narrow; animal medulla is continuous, patterned, and wide.

Pigmentation: Human pigment is denser toward the cuticle; animal pigment is denser toward the medulla.

Scheme of Evaluation

Section	Marking Criteria	Marks
Part A	Full marks for accurate one-sentence or one-word answers. Partial marks (1) for incomplete but correct concepts.	20
Part B	**1 Mark:** Definition; **2 Marks:** Procedure/Mechanism; **1 Mark:** Forensic	20

Section	Marking Criteria	Marks
	Significance.	
Part C	**2 Marks:** Introduction; **5 Marks:** Technical details/Comparison factors; **3 Marks:** Collection/Preservation/Significance.	10
Total		**50**